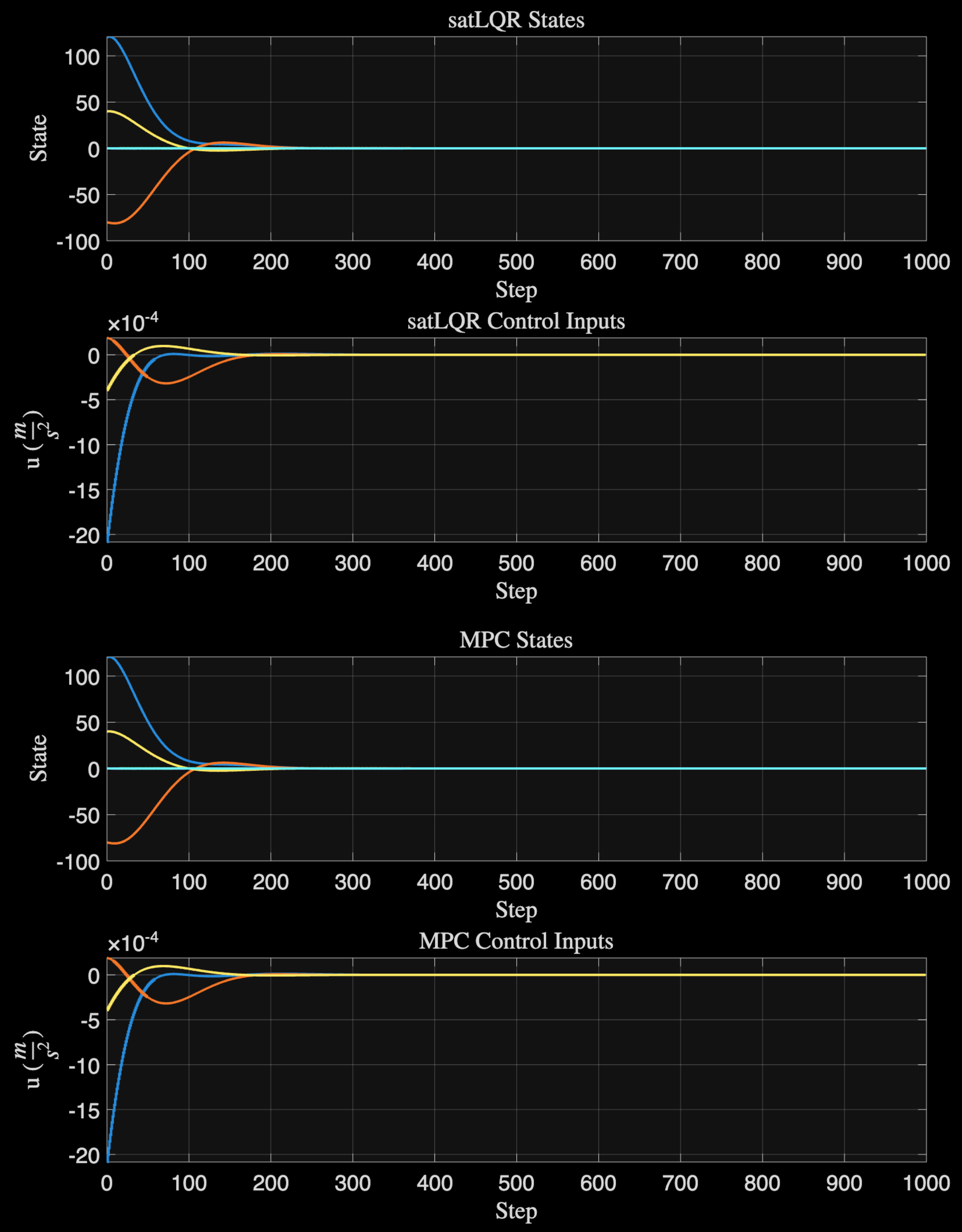
“Fair” Comparison

Trajectories & Trade-offs

- Take the terminal cost to be $\mathbf{diag}(P_f)$ from dlqr
- Use horizon >> time to settle
- Use Q and R with R >> Q s.t. LQR is stabilizing with constraints
- Then simulate without constraints

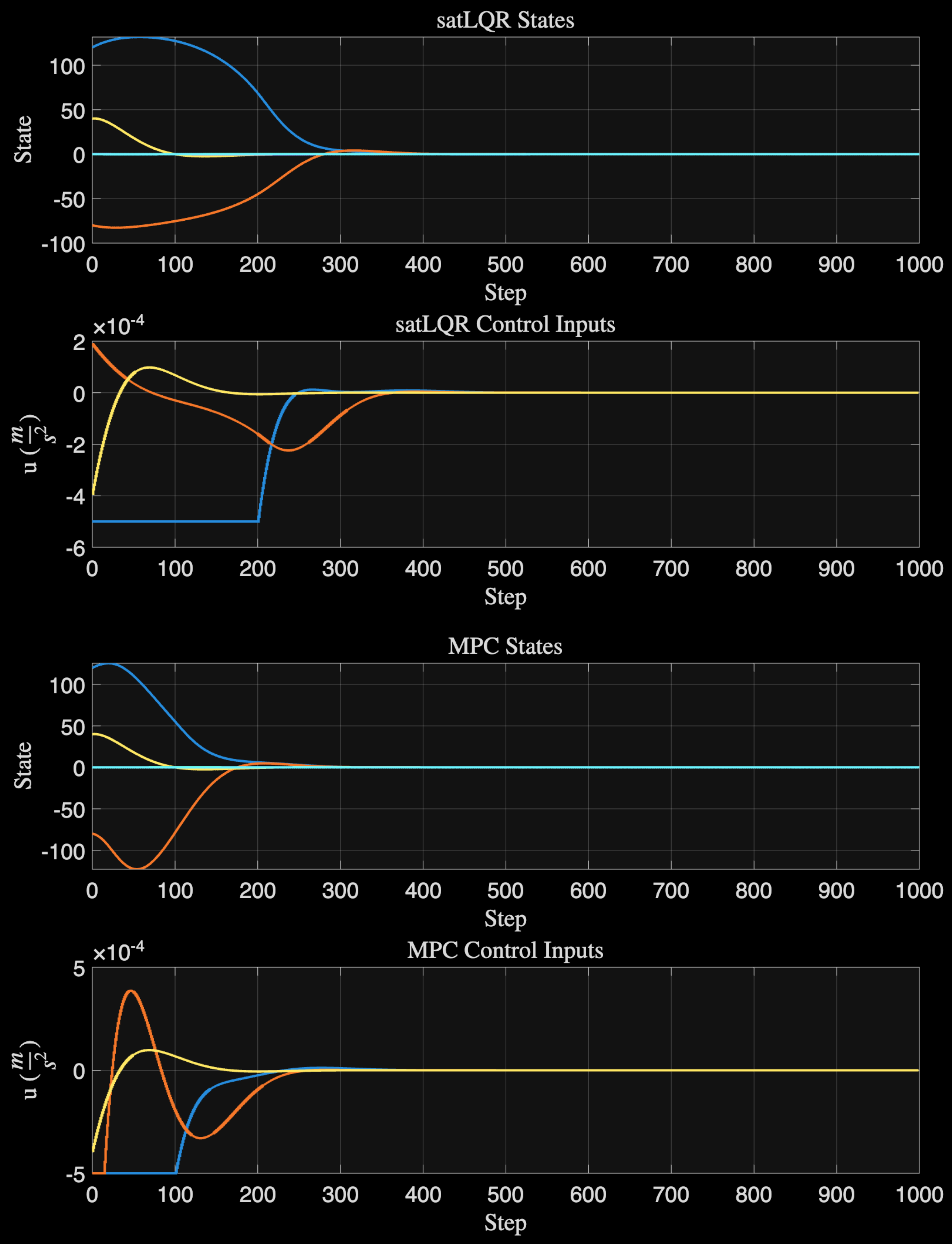
- **Clearly they are the same**



“Fair” Comparison

Trajectories & Trade-offs

- **Now just add the constraint**
  - MPCv2 fuel use → (sum |u|): 0.129735
  - LQR fuel use → (sum |u|): 0.156989
  - MPCv2 lets states grow larger but uses less fuel with the constraint



“Fair” Comparison

Trajectories & Trade-offs

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  - MPCv2 lets states grow larger but uses less fuel with the constraint
- And allowing infeasibility with Nt = 5 gives **5x decrease in fuel** use to 0.0229219
  - (The signal looked more like the economic MPC from before)